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Effects of a Bottom Substrate on Emission Properties of a Photonic Crystal Nanolaser

Se-Heon Kim, Min-Kyo Seo, Ju-Young Kim, and Yong-Hee Lee
Department of physics, Korea Advanced Institute of Science and Technology,
Daejeon 305-701, Korea

Abstract - By choosing a one-wavelength air gap between a two-dimensional photonic crystal slab and a simple InP substrate, over 57% of total radiated power can be collected within a small divergence angle of ±30°

I. Introduction

Two-dimensional (2-D) photonic crystal (PhC) slab structures have been widely investigated as platforms for wavelength-small optical cavities. Intentionally introducing a structural defect in a perfectly periodic PhC, various resonant modes can be designed to the taste of *photonic engineers*. It has been well known that those cavities can have both a very high quality (Q) factor and a very small optical mode volume (V). Since spontaneous emission (SpE) enhancement or, the so called, *Purcell effect* becomes more pronounced with the higher Q/V, one can *control* the SpE rate of an emitter embedded in the cavity. High-efficiency photon sources such as an ultra low-threshold laser and an on-demand single photon source are their possible applications.

Since the first demonstration of nanocavity laser based on the 2-D PhC slab (1), there have been a plenty of improvements on the nanocavity designs. Hwang *et al.* reported the first room-temperature operation of the PhC disk laser, (2) where the PhC slab is placed on an aluminum oxide layer. Park *et al.* demonstrated the first electrically pumped PhC laser (3) based on the monopole mode that has an electric-field null at the cavity center where the current flowing post is placed. By employing InGaAsP-InP quantum well material system, the proposed PhC lasers that have a very high SpE coupling factor β>0.25 can be designed for a telecommunication wavelength of 1.55μm.

Now a technological issue is more placed on the ways to achieve efficient out-coupling into a single mode optical fiber. Recently, Hwang *et al.* demonstrated highly-efficient evanescent coupling between a PhC nanolaser and a *micro*fiber,(4) by which out-coupling efficiency >70% and a record-low threshold of 35μW were achieved. In this case, however, a special cavity mode design should be taken into consideration for good modal overlapping with the propagating mode of the microfiber. In order to achieve a *direct*, free-space launching to the desired optical fiber, one needs to find ways to overcome the strong diffractive tendency of a wavelength-small cavity. Recently, Kim *et al.* proposed the modified hexapole mode that shows very good directional beaming effect. Moreover, it was shown that >80% of the photons generated inside a PhC cavity can be funneled within a small divergence angle of ±30° by employing a high reflective (R>99%) bottom substrate 1λ-below the PhC slab. (5) In this paper, we shall emphasize that even a simple dielectric substrate (such as an InP substrate for InGaAsP-InP material system) can be utilized for fairly good directional beaming of PhC nanolasers.

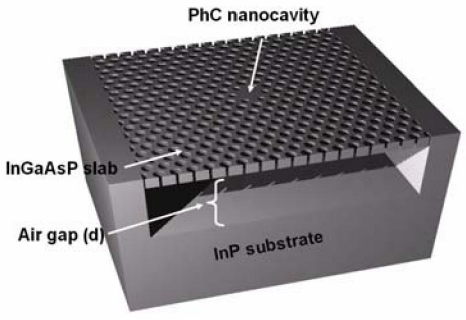


Fig.1. Schematic of a photonic crystal nanolaser which consists of a photonic crystal slab and a simple dielectric substrate.

II. Deformed Hexapole Mode for Directional Far-field Emission

Fig. 1 depicts a schematic of a 2-D slab PhC

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nanolaser. It consists of a PhC slab and a bottom reflector. As a reflector, we have assumed a simple InP substrate whose refractive index is ~3.4. During the wafer growth, one can insert an InGaAsP capping layer between the InP substrate and an InP sacrificial layer. Then, the followed wet-etching process by a diluted HCl will selectively remove the sacrificial layer, which results in an air gap between the PhC slab and the bottom substrate. For detailed fabrication processes of the PhC nanolaser, refer to our review article. (6) It should be noted that a typical thickness of the air gap of 1~2μm is still in the wavelength-scale (peak emission of InGaAsP-InP quantum well ~ 1.55μm). Therefore, one can expect that there must be some kinds of interaction between the cavity mode and the bottom substrate. In order to investigate possible effect, the 3-D finite-difference time-domain method (FDTD) and the near- to far-field transformation formulae based on the Fourier transformation are used. (5)

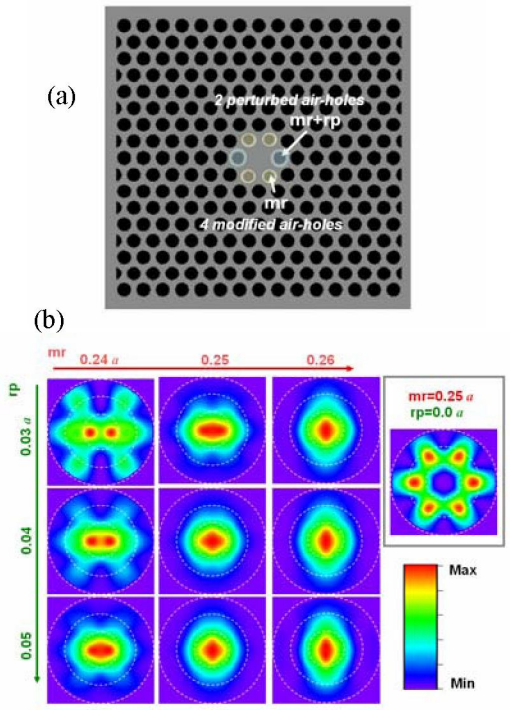


Fig. 2. (a) The deformed hexapole cavity. (b) Calculated far-field emission pattern of the deformed hexapole mode by varying *mr* and *rp*. Here, the slab thickness is chosen to be 0.9*a*.

The PhC nanocavity design that we have employed is shown in Fig. 2(a). Here, the six nearest-neighbor holes are pushed away from the cavity center after reducing their radii from r=0.35a to *mr*, where a is the lattice constant of a triangular PhC. Then, two horizontal air-holes facing with each other are enlarged by *rp*. This secondary modification enables us to break the delicate balance of the six-fold symmetry of the original hexapole mode.(5) For the present, we will forget about the bottom substrate. The resultant far-field emission patterns by varying *mr* and *rp* are shown in Fig. 2(b). Comparing to the far-field pattern of the unperturbed hexapole mode (inset of Fig. 2(b)), gradual transformation from the hexagonally distributed pattern to the directional beaming is clearly noticed. When the perturbed air hole radius, *rp*, is 0.04a and the modified air hole radius, *mr*, is 0.25a, fairly good directional emission can be obtained. In this case, approximately 65% of the total radiated energy falls within the collection angle of ±30°. In spite of the structural perturbation, the calculated Q factors in the range of ~20000 are still not too bad for most applications. However, using a vertically symmetric PhC slab structure, we will lose at least 50% of the downward photons.

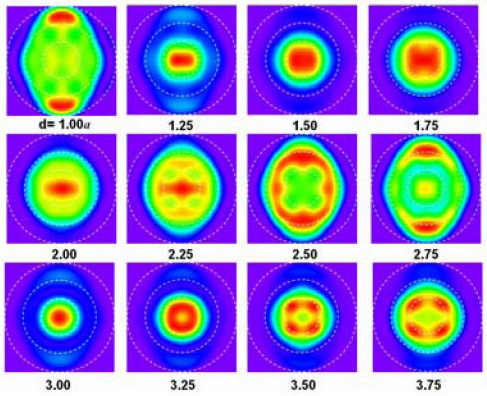


Fig. 3. Calculated far-field patterns of the deformed hexapole mode for various values of the air gap thickness in the presence of a InP substrate

III. Effect of the Bottom Substrate: Unidirectional Beaming

The downward photons can be redirected by introducing a reflector underneath. To investigate effects of the presence of the InP substrate, we have calculated the change of Q factors and far-field radiation patterns as we vary the air-gap distance from 1.00a to 3.75a. Fig. 3 shows far-field radiation patterns for various values of the

air gap thickness. Although the reflectivity of the InP substrate is only ~30%, the interference between the original wave and the reflected wave is easily noticeable. According to our previous analysis, (5) thanks to the 'slab anti-resonance' effect, the vertical enhancement condition becomes rather simple, '$d=\lambda$'. As shown in Fig. 3, the air-gap size of 3.00a, where the good directionality is obtained, is approximately equal to one wavelength (~0.96 λ).

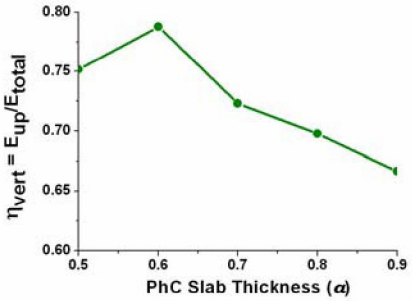


Fig. 4. Estimated vertical coupling efficiency for various values of the photonic crystal slab thickness. Here, mr=0.25a and rp=0.04a have been assumed.

In order to confirm the feasibility of our design, we have estimated the vertical coupling efficiency as follows. During the FDTD time stepping, the Poynting energies that leaks out of the upper (E_{up}) and lower (E_{down}) domain with respect to the PhC slab position are calculated. The vertical coupling efficiency, η_{vert}, is defined as $E_{up}/(E_{up}+E_{down})$. Note that ~1% of additional power loss can take place, horizontally, along the PhC slab, mainly due to the TE-TM mode coupling. (7) The vertical out-coupling efficiencies for various values of the slab thickness (by keeping the λ-air gap condition) are displayed in Fig. 4. η_{vert} varies from 67% to 79%. Then, we have estimated a fraction of angle integrated power (η_{30}) that falls within ±30° from the calculated far-field pattern. The final out-coupling efficiency can be defined as $\varepsilon = \eta_{vert} \times \eta_{30}$.

In fact, the similar vertical beaming effect can be observed when the air gap size is chosen to be half wavelength. It should be noted that, in some applications, for example, for the formation of the current flowing post, (3) smaller air gap is favorable. Fig 5(a). shows the comparison between two far-field patterns obtained by the 1-λ and 0.5-λ air gap, respectively. And angle integrated collection efficiency as a function of the collection angle is shown in Fig. 5(b). One can notice that the 0.5-λ case also shows fairly good directional emission. For the collection efficiencies within ±30°, 1-λ case shows η_{30} of 72.2%, while 0.5-λ case η_{30} of 68.4%. The best final collection efficiency ε is ~57%, when the PhC slab thickness and the air gap size are equal to 0.6a and 1-λ, respectively.

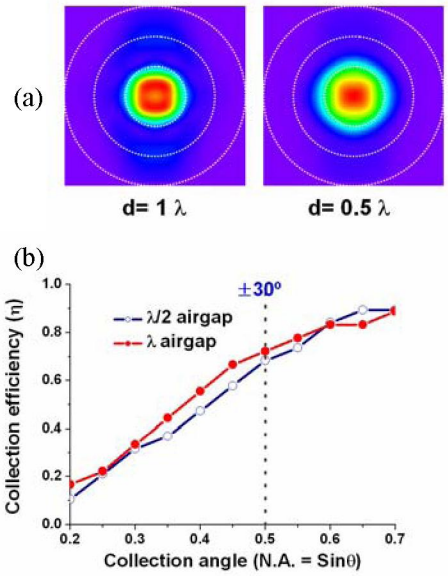


Fig. 5. Far-field emission characteristics of the deformed hexapole cavity with mr=0.25a, rp=0.04a, and t=0.6a. (a) Calculated far-field patterns for two different air gap sizes, 1λ and 0.5λ air gap. (b) Angle integrated collection efficiency.

IV. Conclusion

We have shown that vertical beaming of a wavelength-small PhC nanocavity can be realized by tuning the cavity geometry and placing a simple InP substrate 1-λ below the PhC slab. The estimated vertical coupling efficiency can be as high as 79%, then the final collection efficiency within ±30° can be ~57%. The proposed structure can be applied to on-demand high-efficiency single photon sources and efficient PhC nanolasers. In order to demonstrate the feasibility of the proposed structure, we have fabricated the modified hexapole cavity in an InGaAsP-InP wafer, in which the thickness of the InP sacrificial is 1μm. The detailed experimental result will also be reported in our forthcoming publication.

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